MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

1/ TYPE 8904

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Any
Weight: 4 ounces (113.4 grams) nominal

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Fl	tk	Eb	Ec1	Ec2	Ehk	Ib	Ic1
Unit:	Vac	MHz	sec (min)	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc
	Note 1								
Class AB1:	26.5 ±10%	110	60	2500	---	400	±150	275	2.0
Class A :	26.5 ±10%	110	60	2500	---	400	±150	400	---
TEST CONDITIONS:	26.5	---	240	1000	Adj	300	0	150	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Pg2	Pp	T (anode core and seals)	Cooling	Altitude
Unit:	W	W	°C	---	Ft
				Note 2	
Class AB1:	8.0	350	250	---	10,000
Class A :	8.0	350	250	---	10,000
TEST CONDITIONS:	---	---	---	Note 3	---

GENERAL:

Qualification - Required

Special requirement - See note 19

1/ Formerly designated type 8904/4CX350FJ

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	18					
1256	Electrode current (screen)	---		Ic2	-7.0	+3.0	mAdc
1261	Electrode voltage (grid)	---	Eb = 2,200 Vdc; Ec2 = 400 Vdc; Ec1/Ib = 55 mAdc	Ec1	-19.0	-26.0	Vdc
1266	Total grid current	4	Eb = 2,500 Vdc; Ib = 140 mAdc	Ic1	---	-25.0	µAdc
1266	Primary grid emission (control)	---	Pg1 = 0.5 W; t = 15; anode and screen grid floating	Isg1	---	-25.0	µAdc
1266	Primary grid emission (screen)	---	Ec1 = 0; t = 15; Pg2 = 8.0 W; anode floating	Isg2	---	-250.0	µAdc
1301	Heater current	---		If	0.50	0.81	Aac
1231	Pulsing emission (1)	5,6	etd = eb = ec2 = ec1 = 400 v; Ef = 23.8 Vac	is	20	---	a
1231	Pulsing emission (2)	5,6	eb = ec2 = ec1 = etd/is = 30 a	etd	---	400	v
1372	Current division (long pulse)	---	Eb = 500 Vdc; Ec2 = 400 Vdc; Ec1 = -100 Vdc; prr = 11±1 pps; tp = 4,500 us(min); egk/ib = 850 ma	egk ic2	--- ---	+5.0 260	v ma
1366	Interelement leakage resistance, cold	---	Ef = 0 (30 minutes, min); Rs = 2.5 Meg:	---	---	---	---
			E=100 Vdc; g1 neg	Rg1k	50	---	Meg
			E=500 Vdc; g1 neg	Rg1g2	50	---	Meg
			E=500 Vdc; g2 neg	Rg2p	50	---	Meg
	Quality conformance inspection, part 2	---					
1331	Direct-interelectrode capacitance	---		Cgp	---	0.038	pF
				Cin	20.0	24.0	pF
				Cout	5.6	6.2	pF
1336	Heater-cathode leakage	7	Ehk = +150 Vdc	Ihk	---	100	µAdc
			Ehk = -150 Vdc	Ihk	---	100	µAdc
2204	Linear amplifier power output and distortion (procedure 2.2)	8	Eb = 2,200 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 55 mAdc; Eg1/Ib = 150 mAdc (max) (2 tone); R1=5,000 ±100 ohms; Rg=1,000 ohms(max); Anode tank Q = 10 to 15; t = 180 (max) F = 2 to 10 MHz	Po	250	---	W(PEP)
				3rd IM	-28	---	dB
				5th IM	-31	---	dB
				Ic1	---	50	µAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test (1)	9,10	Group C; linear amplifier power output and distortion (2-tone modulation); ΔIb = 55 mAdc	t	500	---	hrs
---	Life test (1) end points:						
2204	Linear amplifier power output and distortion (procedure 2.2)	---	Eb = 2,200 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 55 mAdc; eg1/Ib = 150 mAdc (max) (2-tone); Rl=5,000 ±100 ohms; Rg=1,000 ohms(max); anode tank Q = 10 to 15; t = 180 (max); F = 2 to 10 MHz	Po 3rd IM 5th IM Ic1	240 -27 -30 ---	--- --- --- 100	W(PEP) dB dB μAdc
1266	Primary grid emission (control)	---	Pg1 = 0.5 W; t = 15; anode and screen grid floating	Isg1	---	-25	μAdc
1266	Primary grid emission (screen)	---	Ec1 = 0; Pg2 = 8.0 W; t = 15; anode floating	Isg2	---	-250	μAdc
1336	Heater-cathode leakage	7	Ehk = +150 Vdc Ehk = -150 Vdc	Ihk Ihk	--- ---	100 100	μAdc μAdc
1366	Interelement leakage resistance, cold	---	Ef = 0 (30 minutes, min); Rs = 2.5 Meg: E=100 Vdc; g1 neg E=500 Vdc; g1 neg E=500 Vdc; g2 neg	--- Rg1k Rg1g2 Rg2p	--- 20 20 20	--- --- --- ---	--- Meg Meg Meg
---	Life test (2)	---	Group C; heater standby; no voltages except Ef = 28.0 Vac	t	500	---	hrs
---	Life test (2) end points:						
2204	Linear amplifier power output and distortion (procedure 2.2)	---	Eb = 2,200 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 55 mAdc; eg1/Ib = 150 mAdc (max) (2-tone); Rl=5,000 ±100 ohms; Rg=1,000 ohms(max); anode tank Q = 10 to 15; t=180 (max); F = 2 to 10 MHz; Ef = 23.8 Vac	Po Ic1	200 ---	--- 100	W(PEP) μAdc
1366	Interelement leakage resistance, cold	---	Ef = 0 (30 minutes, min); Rs = 2.5 Meg: E=100 Vdc; g1 neg E=500 Vdc; g1 neg E=500 Vdc; g2 neg	--- Rg1k Rg1g2 Rg2p	--- 20 20 20	--- --- --- ---	--- Meg Meg Meg

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued						
1011	Humidity	11,16	Ic1 = -30 µAdc (max) under post-test conditions	---	---	---	---
1032	Vibration, mechanical	12,17	Ebb = 2,000 Vdc; Ec1/Ib = 100 mAdc; Rp = 4,900 ohms; 10 G peak (min)	Ep	---	30	Vac
1042	Shock, specified pulse	13,17	Eb = 2,000 Vdc; Ec1 = -100 Vdc; test condition A, except 90 G	---	---	---	---
---	Vibration, mechanical and shock end points:						
1261	Electrode voltage (grid)	---		Ec1	-19.0	-26.0	Vdc
1266	Total grid current	---	Eb = 2,500 Vdc; Ib = 140 mAdc	Ic1	---	-30.0	µAdc
1143	Forced cooling	11,14, 17	Eb = 2,000 Vdc; Ec1/Ib = 175 mAdc; Ic1 = -30 µAdc (max) under post test conditions	T (anode core)	---	250	°C
1155	Coolant-pressure drop versus coolant flow (forced air)	15,17	Eb = 2,000 Vdc; Ec1/Ib = 175 mAdc	---	---	1.3	In.H_2O

NOTES:

1. Maximum tube life may be obtained by adjusting the heater voltage (Ef) in accordance with application. The heater voltage (nominal and derated) shall be maintained within ±5 percent when consistent operation and extended life are factors.

2. At an anode dissipation of 350 watts and an incoming air temperature of 50°C maximum, a minimum airflow of 7.8 cubic feet per minute (cfm) at sea level shall pass through the tube's anode cooler. At this flow rate of 7.8 cfm, the static pressure drop across the tube and socket shown on Drawing 246-JAN is approximately 1.2 inches of water. The pressure drop varies with the amount of escaping air and the shape and construction of the air director. Air cooling of the tube shall be increased with increased incoming air temperature or with increased altitude, or a combination of both. In all cases of operation, a socket which provides forced-air cooling of the base shall be used and maximum seal temperature ratings must not be exceeded. The airflow shall be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them.

3. In all electrical tests involving heater voltage, the socket shown on Drawing 246-JAN shall be used. Forced-air cooling is permitted at the rate of 8.0 cfm maximum for the base and anode, unless otherwise specified in the specific test conditions. A separate source may be used for the base and anode provided neither exceeds 8.0 cfm. Air under standard conditions of temperature and pressure shall be used, or appropriate corrections applied.

4. This test shall be the first test performed at the conclusion of the holding period.

5. The voltages applied to the plate and grids shall not exceed 450 volts. The applied voltage shall have a maximum pulse repetition rate (prr) such that the duty cycle (Du) shall not exceed 0.0002 (0.02 percent) based on the pulse length measured at 50 percent amplitude. The pulse duration (tp) shall not be less than 3 µs at 5 percent of the maximum value, and shall not exceed 2 µs at 50 percent amplitude.

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NOTES: -Continued

6. Method 1231 - pulsing emission (1) or pulsing emission (2) may be performed alternately. However, only one of these tests shall be performed.

7. Measurements shall be taken during an interval of 3 minutes immediately following the required warmup period.

8. Prior to conducting this test, the TUT shall be operated in a static condition (Eb = 2,200 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 55 mAdc) for 3 minutes (minimum) to demonstrate that the Ibo value remains constant (i.e., Ibo = 55 ±5 mAdc). The TUT shall be considered to fail this test if it exhibits instability after 5 minutes of testing. If there is a failure, reprocessing and retesting for stability shall be the only condition permitted for continuing this test.

9. The following measurements shall be made prior to conducting life test (1): Method 1366 - interelement leakage resistance with 50 Meg (minimum) for each of the required measurements, and method 2204 - linear power output and distortion with limits and conditions as specified under quality conformance inspection, part 2.

10. The following measurement shall be made prior to conducting life test (1): Method 1261 - electrode voltage (grid) with the value of Ec1 being read with an accuracy of ±0.1 Vdc and recorded. The TUT's operational stability shall be checked after 96 hours of life testing by repeating the electrode voltage (grid) test, observing the change in Ec1 from the initial reading, and the change in anode current (ΔIb) from the initial value of 55 mAdc. The ΔIb value shall not exceed ±5.0 mAdc and allowable failures shall conform to the group C requirements. In the event of failure after double sampling, the life test lot shall be reprocessed before resubmitting a sample for life testing and the operational stability test.

11. At the conclusion of this test, the TUT shall satisfy the requirements of the total grid current test (method 1266) specified herein under quality conformance inspection, part 1.

12. The TUT shall be mounted in a resonance free jig and vibrated with sinusoidal excitation in each of the three mutually perpendicular planes: (The X-axis shall be defined as normal to a plane drawn through base pins numbered 3 and 7.)

    Survey test: The TUT shall be vibrated for 60 seconds at the frequency which produces maximum vibration output noise voltage in each of the three positions. If at the end of 60 seconds, the vibration noise output is increasing, the vibration shall continue until there is no further increase. There shall be no indication of shorts during this test and the TUT output noise voltage shall not exceed the limit specified herein.

    Cycling test: The frequency shall vary from 28 to 750 to 28 Hz with approximately logarithmic progression, and shall require 6 to 12 minutes to transverse the range, which shall constitute one cycle. The TUT shall be vibrated for one such cycle in each of the three planes.

13. The TUT shall be subjected to six shocks of the specified peak amplitude and duration in each of the three mutually perpendicular directions. Following the impact test, the TUT shall be rejected if it exhibits permanent shorts, or more than one temporary short during the test.

14. The forced cooling test shall be conducted as follows: The base and anode of the TUT shall be cooled by applying an airflow of 7.8 cfm (maximum) from a single source using the infinite baffle system shown on figure 2, or equivalent. Air under standard conditions of temperature and pressure shall be used, or appropriate corrections applied. The anode core temperature shall not exceed the specified limits at the specified test conditions. Temperature shall be measured by means of a thermocouple embedded in the top of the core, adjacent to the cooler. This shall be done by drilling a small hole, shallow enough to prevent loss of TUT vacuum, placing the welded thermocouple junction therein, and then peening the edges of the hole to hold the thermocouple firmly

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in place. In all cases, good electrical continuity between the thermocouple and the metal area in close proximity shall be demonstrated prior to performing the cooling test.

15. This test shall be performed with the TUT operating under the conditions specified for the forced cooling test (see note 14), using the infinite baffle system shown on figure 2, or equivalent. The static pressure drop shall be measured across the TUT and socket.

16. This test shall be performed during the initial production and once each succeeding 3-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of four tubes with an acceptance number of zero, and a second sample of four tubes with a combined acceptance number of one. In the event of failure, this test will be made as a part of quality conformance inspection, part 2, with an AQL of 6.5, inspection level S3. The regular "3-month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

17. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, this test will be made as a part of quality conformance inspection, part 2, with an AQL of 6.5, inspecion level S3. The regular "12-month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

18. The AQL for each test listed under quality conformance inspection, part 1, shall be 0.65, inspection level II.

19. Reclaimed materials shall be utilized to the maximum extent possible as cited in MIL-STD-961 and MIL-STD-962 within the quality limits required by this document and to fulfill compliance with the Resource and Recovery Act of 1976 (Public Law 94-580 dated 21 October 1976).

20. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - CE, MI, SM
Navy -
Air Force - 99
DLA - ES

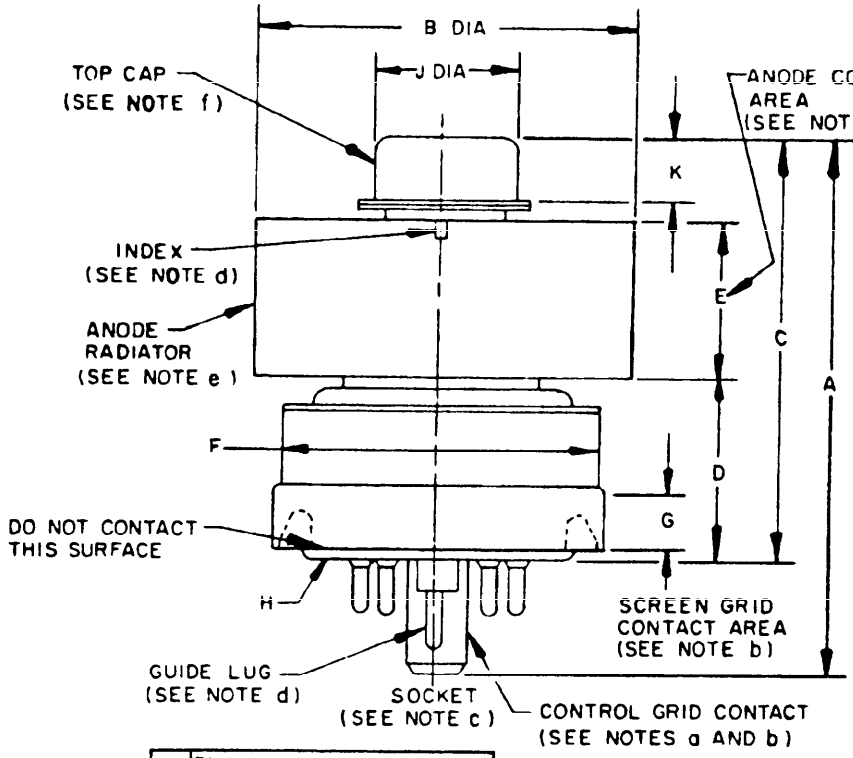
User activities:
Army - AV, ME, SM
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Army - ER

Agent:
DLA - ES

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PIN CONNECTIONS	
Pin No.	Element
1	g2
2	k
3	h
4	k
5	int con
6	k
7	h
8	k
center post	g1
radiator & cap	a
base ring	g2

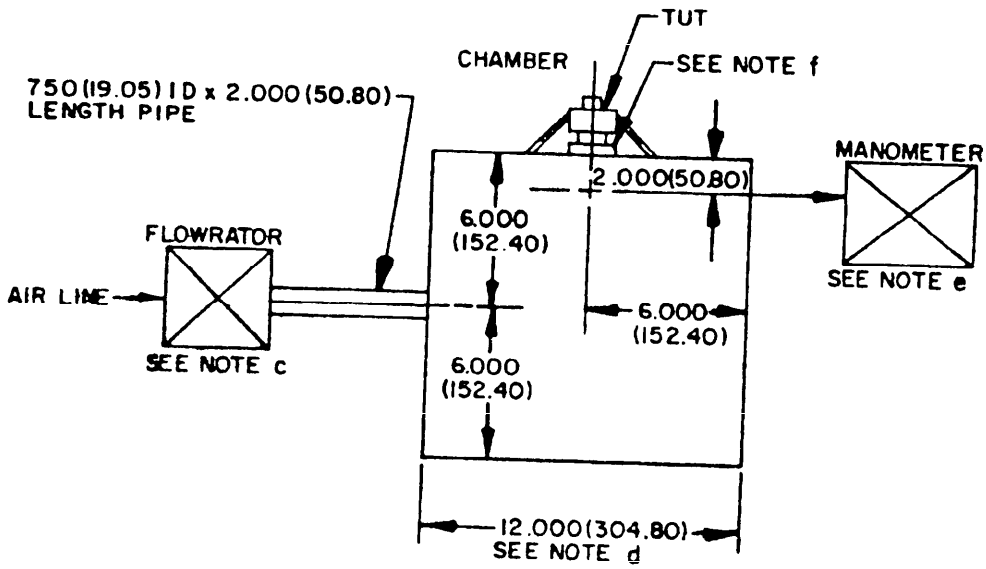
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.324 (59.03)	2.464 (62.59)
C	1.810 (45.97)	1.910 (48.51)
Quality conformance inspection, part 3 (SEE NOTE f7)		
B	1.610 (40.89)	1.640 (41.66)
D	.750 (19.05)	.810 (20.57)
E	.710 (18.03)	.790 (20.07)
F		1.406 (35.71)
G	.187 (4.75)	
H	Base: B8-236 (see note a)	
J	.559 (14.20)	.573 (14.55)
K	.240 (6.10)	

NOTES:

a. Pin alinement shall be checked by means of gage GB8-3. Dimensions of control-grid contact shall be inspected by means of gages specified on Drawing 246-JAN and shall be quality conformance inspection, part 2.
b. Alinement of anode, screen-grid, and control-grid contact surfaces shall be determined by means of gage specified on Drawing 168-JAN. Quality conformance inspection, part 2, shall apply.
c. Air-system socket shall be as specified on Drawing 246-JAN, EIMAC SK-600, or equal.
d. Location of guide lug of control-grid contact shall be referenced by a notch or arrow on the anode radiator in position shown.
e. Anode clamping shall be confined to anode radiator.
f. Top cap outline optional provided it meets requirements of dimensions J and K.

FIGURE 1. Outline drawing of electron tube type 8904.

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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.
c. Fisher Porter flowrator model B4-27-10/77, or equal.
d. 12 inch (304.80 mm) cube inside dimensions, compound sealed.
e. F.W. Dwyer manometer, 0 to 1 inch (25.40 mm) of water (Fisher Scientific Company 11-295-5 draft gage), or equal.
f. Socket specified on drawing 246-JAN.

FIGURE 2 Baffle system.

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